$$\lambda = \frac{h}{m\upsilon} = \frac{h}{\sqrt{2emV}} \qquad (8.12)$$

Substituting the known values in the above equation, we get

$$\lambda = \frac{6.626\times10^{-34}}{\sqrt{2V\times1.6\times10^{-19}\times9.11\times10^{-31}}}$$
$$= \frac{12.27\times10^{-10}}{\sqrt{V}}\,\text{m} \quad \text{(or)}$$
$$\lambda = \frac{12.27}{\sqrt{V}}\,\text{Å}$$

For example, if an electron is accelerated through a potential difference of 100V, then its de Broglie wavelength is 1.227 Å.

Since the kinetic energy of the electron, $K = eV$, then the de Broglie wavelength associated with electron can be also written as

$$\lambda = \frac{h}{\sqrt{2mK}} \qquad (8.13)$$

8.3.4 Davisson – Germer experiment

Louis de Broglie hypothesis of matter waves was experimentally confirmed by Clinton Davisson and Lester Germer in 1927. They demonstrated that electron beams are diffracted when they fall on crystalline solids. Since crystal can act as a three-dimensional diffraction grating for matter waves, the electron waves incident on crystals are diffracted off in certain specific directions. Figure 8.17 shows a schematic representation of the apparatus for the experiment.

The filament F is heated by a low tension (L.T.) battery. Electrons are emitted from the hot filament by thermionic emission. They are then accelerated due to the potential difference between the filament and the anode aluminium cylinder by a high tension (H.T.) battery. Electron beam is collimated by using two thin aluminium diaphragms and is allowed to strike a single crystal of Nickel.

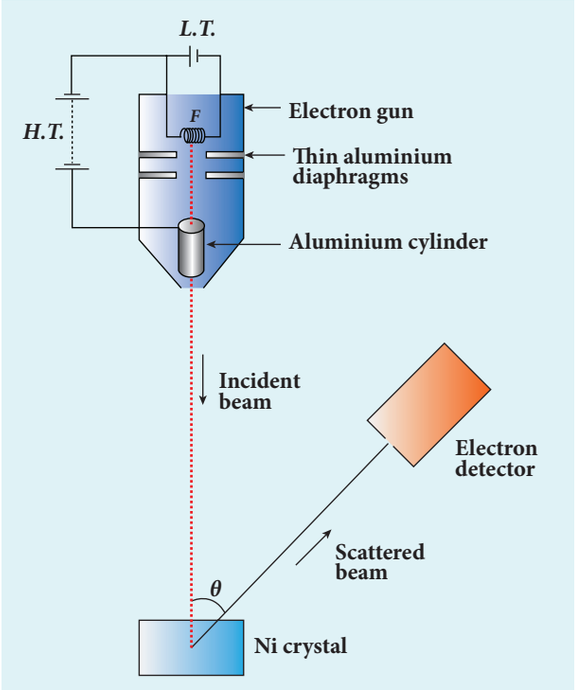


Figure 8.17 Experimental set up of Davisson – Germer experiment

The electrons scattered by Ni atoms in different directions are received by the electron detector which measures the intensity of scattered electron beam. The detector is capable of rotation in the plane of the paper so that the angle θ between the incident beam and the scattered beam can be changed at our will. The intensity of the scattered electron beam is measured as a function of the angle θ.

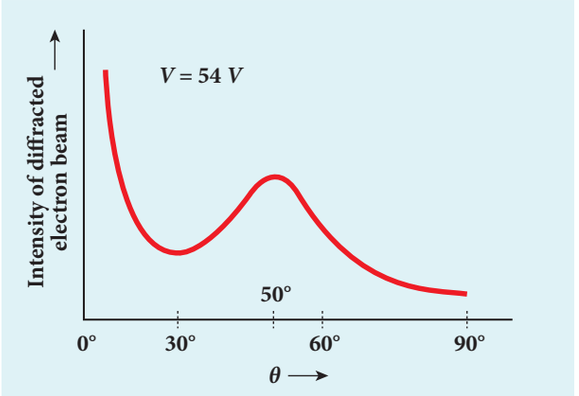


Figure 8.18 Variation of intensity of diffracted electron beam with the angle θ

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